embedUR

Linux Assingment-2

Name: Saravanan D
Collage: VIT University

1) Use the appropriate command to list all files larger than 1 MB in the current directory and save the output to a file.

find . -type f -size +1M > big_file.txt
cat big_file.txt

```
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

vboxuser@Ubuntu:~$ find . -type f -size +1M > big_file.txt
vboxuser@Ubuntu:~$ cat big_file.txt
./.cache/mesa_shader_cache_db/index
./.cache/tracker3/files/http%3A%2F%2Ftracker.api.gnome.org%2Fontology%2Fv3%2Ftracker%23Audio.db-wal
./.cache/tracker3/files/meta.db-wal
./.cache/tracker3/files/http%3A%2F%2Ftracker.api.gnome.org%2Fontology%2Fv3%2Ftracker%23Documents.db-wal
./.cache/tracker3/files/http%3A%2F%2Ftracker.api.gnome.org%2Fontology%2Fv3%2Ftracker%23Software.db-wal
./.cache/tracker3/files/http%3A%2F%2Ftracker.api.gnome.org%2Fontology%2Fv3%2Ftracker%23Video.db-wal
./.cache/tracker3/files/http%3A%2F%2Ftracker.api.gnome.org%2Fontology%2Fv3%2Ftracker%23Pictures.db-wal
./.cache/tracker3/files/http%3A%2F%2Ftracker.api.gnome.org%2Fontology%2Fv3%2Ftracker%23FileSystem.db-wal
vboxuser@Ubuntu:~$
```

2) Replace all occurrences of "localhost" with "127.0.0.1" in a configuration file named config.txt, and save the updated file as updated_config.txt.

touch config.txt
nano config.txt
sed ‘s/localhost/127.0.01/g’ config.txt > updated_config.txt
cat updated_config.txt

```
vboxuser@Ubuntu:~$ touch config.txt
vboxuser@Ubuntu:~$ nano config.txt
vboxuser@Ubuntu:~$ sed 's/localhost/127.0.0.1/g' config.txt > updated_config.txt
vboxuser@Ubuntu:~$ cat updated_config.txt
server=127.0.0.1
db_host=127.0.0.1
vboxuser@Ubuntu:~$
```

```
  GNU nano 7.2                                    config.txt
server=localhost
db_host=localhost
```

3) Use the appropriate command to search for lines containing the word "ERROR" in a log file but exclude lines containing "DEBUG". Save the results to a file named filtered_log.txt.

log:

DEBUG: Starting the application initialization.
ERROR: Unable to connect to the database.
INFO: User 'admin' logged in successfully.
DEBUG: Fetching configuration settings from the server.
ERROR: DEBUG - Configuration settings could not be applied.
INFO: Scheduled job 'backup' completed successfully.
DEBUG: Connection to server timed out. Retrying...
ERROR: Failed to fetch data from API endpoint '/users'.
INFO: Maintenance mode activated.
ERROR: DEBUG - Query execution failed due to a syntax error.
DEBUG: Reloading application modules.
ERROR: Missing required parameter in the request.
INFO: Shutting down the system gracefully.
DEBUG: Closing unused network connections.
ERROR: DEBUG - Unexpected server response received.

touch app.log
nano app.log
grep “ERROR” app.log | grep -v “DEBUG” > filtered_log.txt
cat filtered_log.txt

```
vboxuser@Ubuntu:~$ touch app.log
vboxuser@Ubuntu:~$ nano app.log
vboxuser@Ubuntu:~$ grep "ERROR" app.log | grep -v "DEBUG" >filtered_log.txt
vboxuser@Ubuntu:~$ cat filter
filtered_log.txt  filter_log.txt
vboxuser@Ubuntu:~$ cat filtered_log.txt
ERROR: Unable to connect to the database.
ERROR: Failed to fetch data from API endpoint '/users'.
ERROR: Missing required parameter in the request.
vboxuser@Ubuntu:~$
```

4) Write a code to identify the process with the highest memory usage and then terminate it.

ps aux –sort=-%mem
kill -9 2909

```
Feb 13 05:36
vboxuser@Ubuntu: ~
vboxuser@Ubuntu:~$ pwd
/home/vboxuser
vboxuser@Ubuntu:~$ ps aux --sort=-%mem
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
vboxuser    3482  9.0 14.0 3276524 681148 ?      Sl   05:12   2:03 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    2909 11.1 11.7 11867468 568948 ?     Sl   05:11   2:33 /snap/firefox/7766/usr/lib/firefox/firefox
vboxuser    1881  2.6  8.6 4019212 419988 ?      Ssl  04:38   1:29 /usr/bin/gnome-shell
vboxuser    2542  0.0  4.6 1583696 224696 ?      Ssl  04:38   0:02 /usr/libexec/xdg-desktop-portal-gnome
vboxuser    3131  0.8  3.1 2634880 151876 ?      Sl   05:11   0:11 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    3087  0.1  2.1 2599792 104144 ?      Sl   05:11   0:02 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    3680  1.0  2.0 2590396 96868 ?       Sl   05:12   0:14 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    3903  0.0  1.4 2561056 69696 ?       Sl   05:13   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    3937  0.0  1.4 2561056 69692 ?       Sl   05:13   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    3687  0.2  1.4 2560308 69548 ?       Sl   05:12   0:03 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    4380  0.4  1.2 2805780 61212 ?       Sl   05:31   0:00 gjs /usr/share/gnome-shell/extensions/ding@rastersoft
vboxuser    2049  0.0  1.2 897404 61080 ?        Sl   04:38   0:00 /usr/libexec/evolution-data-server/evolution-alarm-no
vboxuser    2722  0.0  1.1 554948 55244 ?        Ssl  04:40   0:02 /usr/libexec/gnome-terminal-server
vboxuser    3672  0.0  1.0 584428 51468 ?        Sl   05:12   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    3098  0.0  0.9 582380 47392 ?        Sl   05:11   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    1988  0.0  0.9 592744 43564 ?        Ssl  04:38   0:00 /usr/libexec/evolution-source-registry
root        4259  0.2  0.8 480652 42236 ?        Ssl  05:23   0:01 /usr/libexec/fwupd/fwupd
root         670  0.1  0.8 1849924 40556 ?       Ssl  04:37   0:04 /usr/lib/snapd/snapd
vboxuser    3070  0.0  0.8 449380 40200 ?        Sl   05:11   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    3066  0.0  0.7 434940 35520 ?        S    05:11   0:00 /snap/firefox/7766/usr/lib/firefox/firefox -contentpr
vboxuser    2657  0.0  0.6 642488 33708 ?        Sl   04:39   0:00 /usr/bin/update-notifier
vboxuser    2349  0.0  0.6 429632 31136 ?        Sl   04:38   0:00 /snap/snapd-desktop-integration/343/usr/bin/snapd-des
vboxuser    2315  0.0  0.6 751592 30932 ?        Ssl  04:38   0:00 /usr/libexec/evolution-addressbook-factory
vboxuser    2209  0.1  0.6 420968 29220 ?        Sl   04:38   0:03 /usr/libexec/ibus-extension-gtk3
vboxuser    2477  0.0  0.5 2592980 27608 ?       Sl   04:38   0:01 /usr/bin/gjs -m /usr/share/gnome-shell/org.gnome.Scre
vboxuser    2000  0.0  0.5 2519336 27572 ?       Sl   04:38   0:00 /usr/bin/gjs -m /usr/share/gnome-shell/org.gnome.Shel
vboxuser    2526  0.0  0.5 734592 27124 ?        SNsl 04:38   0:01 /usr/libexec/tracker-miner-fs-3
vboxuser    2017  0.0  0.5 592900 26328 ?        Ssl  04:38   0:00 /usr/libexec/gsd-power
vboxuser    2564  0.0  0.5 417244 25396 ?        Ssl  04:38   0:00 /usr/libexec/xdg-desktop-portal-gtk
```

```
Feb 13 05:36
vboxuser@Ubuntu: ~
root          85  0.0  0.0      0     0 ?        I    04:37   0:01 [kworker/u10:2-events_power_efficient]
root          90  0.0  0.0      0     0 ?        I<   04:37   0:00 [kworker/R-charger_manager]
root         140  0.0  0.0      0     0 ?        S    04:37   0:00 [scsi_eh_2]
root         141  0.0  0.0      0     0 ?        I<   04:37   0:00 [kworker/R-scsi_tmf_2]
root         194  0.1  0.0      0     0 ?        S    04:37   0:04 [jbd2/sda2-8]
root         195  0.0  0.0      0     0 ?        I<   04:37   0:00 [kworker/R-ext4-rsv-conversion]
root         357  0.0  0.0      0     0 ?        S    04:37   0:00 [psimon]
root         571  0.0  0.0      0     0 ?        S    04:37   0:00 [irq/18-vmwgfx]
root         577  0.0  0.0      0     0 ?        I<   04:37   0:00 [kworker/R-ttm]
root        1112  0.0  0.0      0     0 ?        S    04:37   0:00 [psimon]
root        1583  0.0  0.0      0     0 ?        I<   04:37   0:00 [kworker/u13:1]
root        2759  0.0  0.0      0     0 ?        I<   04:47   0:00 [kworker/u12:2-ttm]
root        2766  0.0  0.0      0     0 ?        I<   04:47   0:00 [kworker/0:2H-kblockd]
root        2767  0.1  0.0      0     0 ?        I    04:47   0:03 [kworker/u10:0-kvfree_rcu_reclaim]
root        2770  0.0  0.0      0     0 ?        I    04:48   0:00 [kworker/u9:0-events_power_efficient]
root        2773  0.0  0.0      0     0 ?        I    04:50   0:00 [kworker/1:1-cgroup_release]
root        2792  0.0  0.0      0     0 ?        I    04:50   0:01 [kworker/u10:3-events_power_efficient]
root        2822  0.0  0.0      0     0 ?        I<   05:04   0:00 [kworker/u12:0-ttm]
root        3053  0.1  0.0      0     0 ?        I    05:11   0:01 [kworker/u9:1-kvfree_rcu_reclaim]
root        3880  0.0  0.0      0     0 ?        I    05:12   0:00 [kworker/0:0-events]
root        4149  0.3  0.0      0     0 ?        I    05:16   0:03 [kworker/1:2-events]
root        4161  0.1  0.0      0     0 ?        I<   05:17   0:01 [kworker/1:1H-kblockd]
root        4190  0.0  0.0      0     0 ?        I    05:20   0:00 [kworker/0:2-cgroup_free]
root        4206  0.0  0.0      0     0 ?        I    05:20   0:00 [kworker/u10:4-events_power_efficient]
root        4237  0.3  0.0      0     0 ?        I<   05:22   0:02 [kworker/0:0H-kblockd]
root        4239  0.0  0.0      0     0 ?        I<   05:22   0:00 [kworker/u13:2]
root        4315  0.0  0.0      0     0 ?        I    05:27   0:00 [kworker/u9:2-flush-8:0]
root        4317  0.0  0.0      0     0 ?        I<   05:28   0:00 [kworker/1:2H-kblockd]
root        4334  0.0  0.0      0     0 ?        I    05:30   0:00 [kworker/1:0-cgroup_free]
root        4336  0.0  0.0      0     0 ?        I    05:30   0:00 [kworker/u10:1-events_unbound]
vboxuser@Ubuntu:~$ kill -9 2909
vboxuser@Ubuntu:~$
```

5) Use the networking tool command and print all the gateway available in a sorted manner

ip route
ip | route | grep default | awk ‘{print $3}’ |sort

```
vboxuser@Ubuntu:~$ ip route
default via 10.0.2.2 dev enp0s3 proto dhcp src 10.0.2.15 metric 100
10.0.2.0/24 dev enp0s3 proto kernel scope link src 10.0.2.15 metric 100
vboxuser@Ubuntu:~$ ip route | grep default | awk '{print $3}' |sort
10.0.2.2
vboxuser@Ubuntu:~$
```